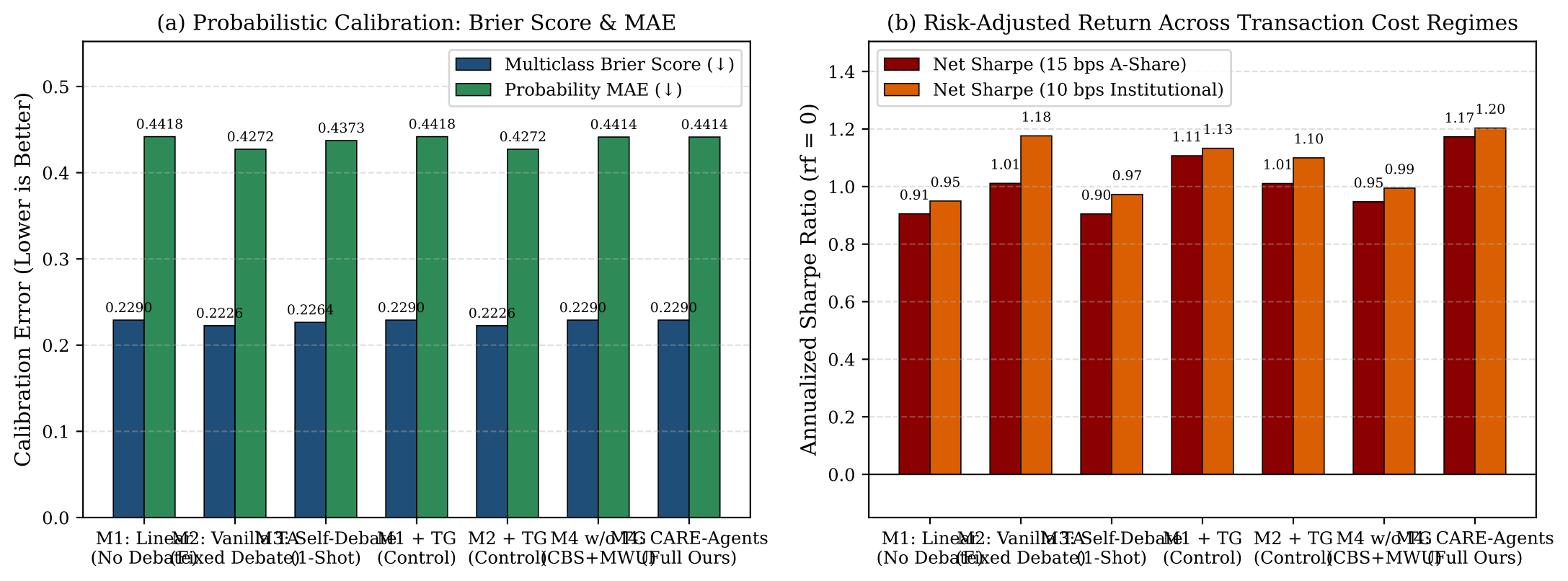
(a) Probabilistic Calibration: Brier Score & MAE
Multiclass Brier Score (↓)
Probability MAE (↓)
Calibration Error (Lower is Better)
0.0
0.1
0.2
0.3
0.4
0.5
0.4418
0.4272
0.4373
0.4418
0.4272
0.4414
0.4414
0.2290
0.2226
0.2264
0.2290
0.2226
0.2290
0.2290
M1: Linear (No Debate)
M2: Vanilla (Fixed Debate)
M3: A-Self-Debate (1-Shot)
M1 + TG (Control)
M2 + TG (Control)
M4 w/o MWU (CBS)
M4: CARE-Agents (Full Ours)
(b) Risk-Adjusted Return Across Transaction Cost Regimes
Net Sharpe (15 bps A-Share)
Net Sharpe (10 bps Institutional)
Annualized Sharpe Ratio (rf = 0)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0.91
0.95
1.01
1.18
0.90
0.97
1.11
1.13
1.01
1.10
0.95
0.99
1.17
1.20
M1: Linear (No Debate)
M2: Vanilla (Fixed Debate)
M3: A-Self-Debate (1-Shot)
M1 + TG (Control)
M2 + TG (Control)
M4 w/o MWU (CBS)
M4: CARE-Agents (Full Ours)